

THE BIG IDEA IN ONE SENTENCE

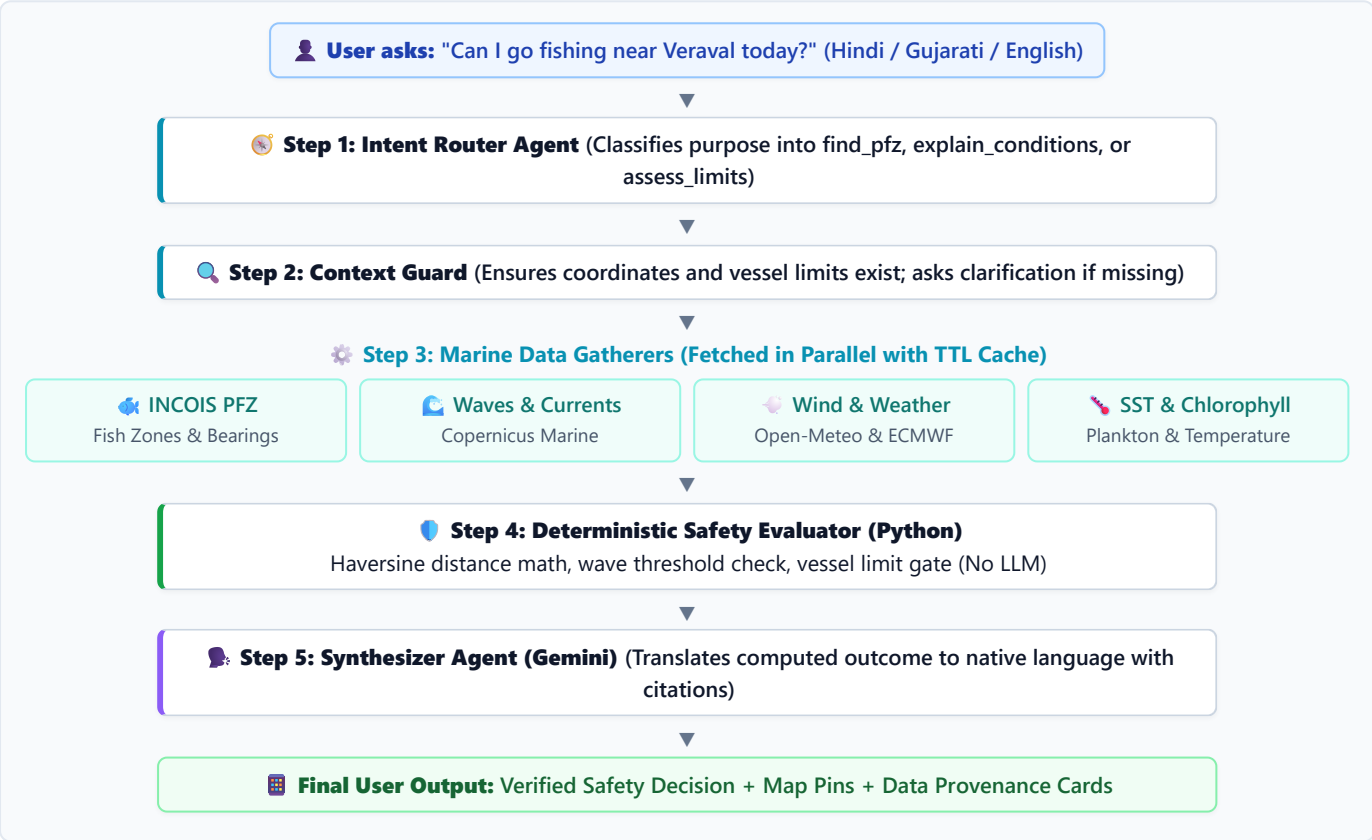
"ORCA is not a ChatGPT clone that guesses answers. It is a 'Deterministic Brain with an AI Mouth.'"

Most AI chatbots hallucinate numbers or fabricate coordinates. In marine operations, wrong numbers cost lives. That's why **ORCA never lets an LLM do math or invent coordinates.**

- **Python handles the facts:** Exact GPS distances, wind speeds, wave heights, and government safety rules.
- **The AI Agent handles communication:** Understanding the fisherman's question (in English, Hindi, Gujarati, etc.) and explaining the computed safety advice in clear, human language.

The 5-Step Agent Flow

Here is what happens in the 6–7 seconds between a user asking a question and receiving an answer:



Deep Dive: The 5 Agents in the System

1. The Intent Router Agent

THE AIR TRAFFIC CONTROLLER

- **What it does:** Reads the user's message and figures out what they actually want.
- **Examples of intents:** find_pfz ("Where are the fish?"), explain_conditions ("How are the waves and wind today?"), assess_limits ("Is it safe for my 10-meter trawler?").
- **Why it's smart:** Powered by **Google Gemini 2.5/3.7 Flash** to understand regional nuances in Indian coastal languages (Hindi, Gujarati, Tamil, Marathi, Bengali).
- **Failsafe:** If Gemini is rate-limited or offline, an internal **Deterministic Keyword Router** takes over instantly. The system never crashes or goes unavailable.

2. 🔍 The Capability Guard

THE SAFETY INSPECTOR

- What it does:** Makes sure we have sufficient factual parameters before giving advisory advice.
- Why it matters:** If a user asks *"Is it safe to go out?"* without providing their GPS location or vessel size, a generic chatbot might dangerously guess.
- ORCA's Guard blocks that:** It verifies coordinates, time window, and vessel thresholds. If missing, it politely asks for clarification instead of gambling with safety.

3. 🌐 The Data Gatherers

THE RESEARCH SQUAD

- What they do:** Fetch official real-time data from 7 trusted scientific providers in parallel:
 - INCOIS (Govt of India):** Live coastal Potential Fishing Zone (PFZ) advisory bulletins.
 - Copernicus Marine (EU):** Sea surface temperature (SST), chlorophyll (plankton concentration), currents, and sea levels.
 - Open-Meteo & ECMWF:** Real-time wave heights, swell periods, wind speeds, and gusts.
- Speed trick:** Employs an **in-memory TTL cache**. If another boat from the same harbour asks 1 minute later, ORCA returns verified data instantly without flooding government APIs.

4. 🛡️ The Deterministic Safety Evaluator

THE MATH ENGINE

- What it does:** Pure Python algorithms (100% deterministic, zero LLM hallucination risk).
- Math performed:**
 - Haversine Formula:** Calculates true nautical distance (in km) and bearing (compass degrees) between harbour and fishing zone.
 - Threshold Check:** Compares live wave height (e.g. 2.4m) against vessel limits (e.g. 2.0m).
 - Safety Veto Gate:** If limits are breached or an active official warning exists, it triggers `EXCEEDS_CONFIGURED_LIMITS`.

5. 🗣️ The Synthesizer Agent

THE TRANSLATOR & EXPLAINER

- What it does:** Takes the validated numbers, safety outcome, and provenance references, and crafts an empathetic explanation in the user's chosen language.
- Enforced safety rules:** The LLM is strictly prohibited from altering calculated numbers or overriding deterministic safety gates. It cites official validity dates on all advisories.

💡 Quick Summary Cheat Sheet (For Presentations & Demos)

Feature	Standard AI Chatbot	ORCA Decision-Support System
GPS Distance & Bearing	Guessed by LLM (often inaccurate by 50+ km)	Calculated via Haversine Python algorithms
Marine Feeds	Training data (months or years old)	Real-time live feeds (INCOIS, Copernicus, Open-Meteo)
Language Support	English-only or poor translation	Native multilingual (Hindi, Gujarati, Tamil, etc.)
Reliability	Hallucinates when missing info	Deterministic safety gates block unsafe advice
Internet / API Drop	Fails completely	Fallback router + cached stale fallback